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The British Tertiary Igneous Province: Palaeomagnetism of the Vaternish Dyke Swarm on North Skye, Scotland

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Summary

We present here the results of a palaeomagnetic investigation of 87 dykes from the north-western area of the regional linear dyke-swarm of Skye, NW Scotland, as a further contribution to studies of the British Tertiary igneous province. Twenty per cent of the dykes are normally magnetized, as with the north Mull dyke swarm. Reversely magnetized dykes with very low inclinations are present, as they are in some previous studies of the province. There are groups of closely adjacent dykes in which the individual dykes of any one group have very similar directions of magnetization—'palaeomagnetic dyke groups'.

The overall distribution of dyke-mean magnetizations is very far from being Fisherian. A mean pole position is not considered to be a useful expression of the results.

1. Introduction

This paper concerns a suite of 87 dykes of lower Tertiary age from the Vaternish peninsula at the north of the Isle of Skye, Scotland. It is a further contribution to intensive palaeomagnetic and geochronological studies of the British Tertiary igneous province. Previous papers in the series containing references to earlier work, have been published by Ade-Hall *et al.* (1972), Purdy *et al.* (1972), Wilson, Dagley & Ade-Hall (1972) and Wilson (1970).

2. Geological setting of the dykes

The dykes of the Vaternish peninsula are intruded into the Ramascaig group of lavas (Anderson & Dunham 1966, p. 82), which was examined as lava sequences J and K by Wilson *et al.* (1972). The group of 87 dykes outcrops between Ardmore Point and Vaternish Point on the west side of the Vaternish peninsula (Fig. 1). The dykes are members of the regional linear dyke-swarm of Skye (Speight 1972). The dykes of this swarm are NW to NNW trending and of basaltic composition. An axis of high density of the swarm passes north–north–westwards from the Cuillin Hills centre, which forms part of the central intrusive complex, through Vaternish peninsula to Harris (inset, Fig. 1). At an axial zone on Vaternish, 2·5 km south of Vaternish Point, the density over a short distance reaches a maximum of 60 dykes per km, producing 10 per cent crustal extension. At Ardmore Point, 4 km south of the axis, the intensity falls to 5 dykes per km (*ca* 1·5 per cent dilation) and at Vaternish Point to 10 dykes per km (*ca* 2·5 per cent dilation). The dykes of the Vaternish peninsula

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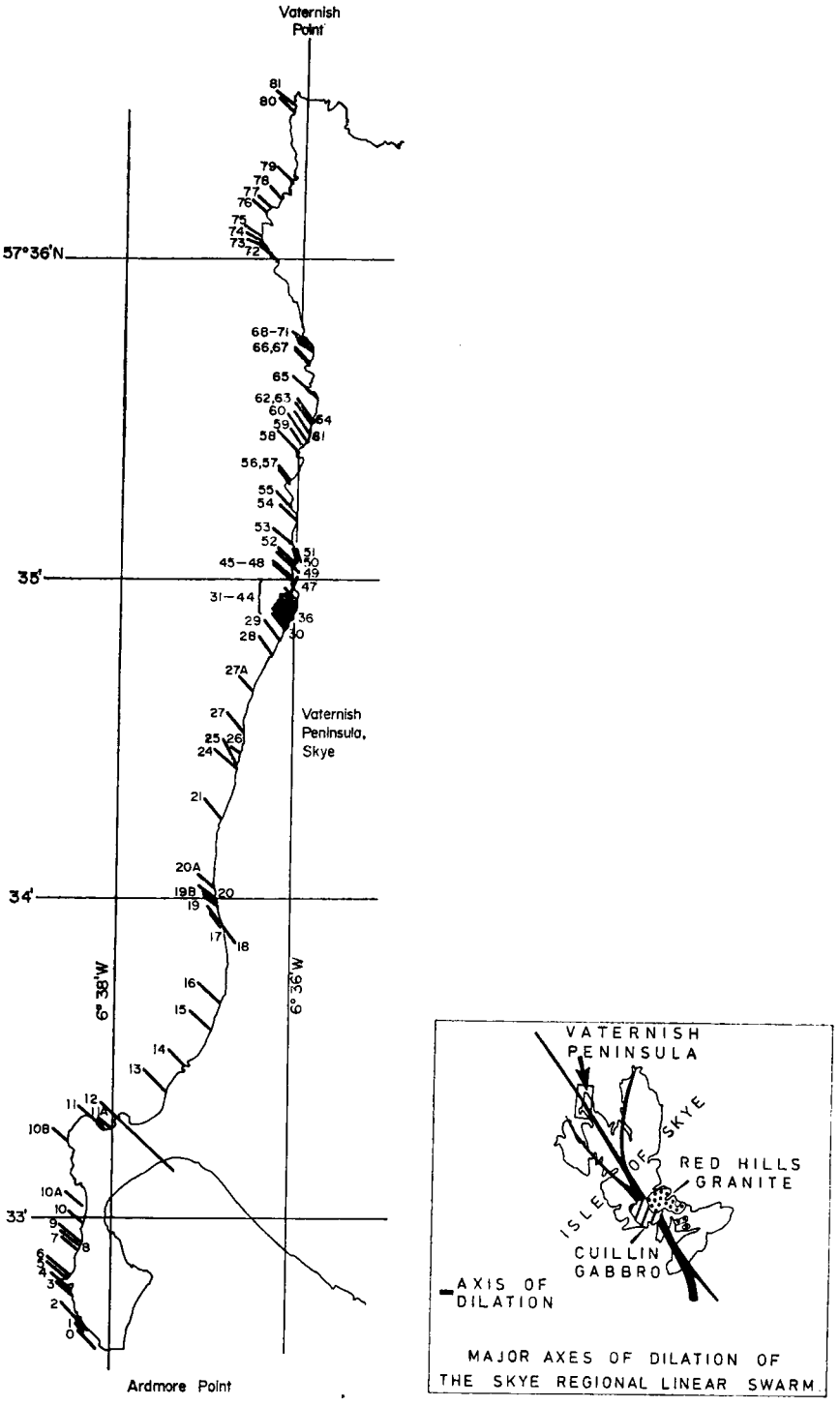


FIG. 1. Locations of the dykes in the Vaternish peninsula. Dykes numbered on the seaward side have reversed or shallow reversed magnetizations. Dykes numbered on the landward side have normal magnetizations. Dykes 19A, 22 and 23 were not sampled. Inset map shows the location of the sampled dykes in relation to the dilation axes of the Skye Regional Linear Dyke Swarm.

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range up to 5 m thickness and average at just less than 2 m. Their trend is between 300° and 350° east of north, the arithmetic average for the group being 325°. One of the advantages of sampling here is that the dykes are far removed from the metamorphic effects of the central intrusive complex. They have experienced zone B hydrothermal alteration (Ade-Hall, Palmer & Hubbard (1971)) conditions, based on curves of saturation magnetization *vs* temperature, with reheating to not more than 100 °C in the presence of abundant groundwater.

3. Field and laboratory procedure

The dykes of Vaternish are labelled by the letter 'I' in our Tertiary collection.

At least four separately oriented 25 mm diameter drill-cores were taken from each of 87 dykes, for palaeomagnetic and other measurements. Orientation was by means of sun-sighting, and in a few instances by moon-sighting. Each core was sliced into 25 mm length specimens. One specimen from each core was demagnetized in 5 mT (1 milliTesla = 10 Gauss) steps up to 25 mT initially. Many were further demagnetized to higher fields, up to 150 mT after inspection of the first results. A mean direction of natural magnetization was then determined for each dyke according to the procedure of Dagley & Ade-Hall (1970). Two dykes with α_{95} greater than 10° were rejected from analysis. The mean α_{95} for the remaining 85 dykes was 3·36°.

No correction for tilt of the dykes could be made. Although it will not exceed several degrees, it is difficult to estimate the exact amount of tilting of the Vaternish region subsequent to dyke-injection. The dips of the lavas of northern Skye are thought in many cases to be the original dips (Anderson & Dunham 1966, p. 176) and are not the consequence of tectonic uplift.

The intrusion of the Vaternish dyke assemblage is not restricted to a single period. A process of continual gravitational downwarping of the lava-pile during its accretion must have affected the dips of all but the post-lava dykes. The earliest dykes would have suffered most tilting. Further complications arise from the diversity of sources of the lavas, emanating as is believed (Anderson & Dunham 1966, p. 79) from a number of reservoirs. The downwarp of the pile in this case was not of constant orientation, but towards the successive foci of eruption.

The evidence of tilting of the region yielded by the dykes is inconclusive. Distribution of amygdales in the bodies of individual dykes is in some cases uniform throughout the dyke, in some cases concentrated towards the hanging-wall and in other cases in central bands. In the first and last cases a post-intrusive tilt of an originally vertical dyke is indicated. In the second case, which occurs among dykes dipping towards both north-east and south-west, the evidence points to a primary dip, though the amount may not be equivalent to the present dip.

As a result of all the variable influences affecting the dips of the dykes, it is unfortunately impossible to calculate for any individual dyke the degree and direction of post-intrusive tilt.

4. Results of palaeomagnetic measurements

Fig. 2 and the table in the Appendix contain the overall dyke-mean palaeomagnetic results.

4.1. *Normally magnetized dykes*

Unlike the British Tertiary lavas, but like the Mull dyke swarm, the Vaternish samples contains 20 per cent of normally magnetized dykes. However, unlike the case of the Mull swarm (Fig. 8, Ade-Hall *et al.* 1972), these normal directions have a considerable spread. In Fig. 3 where the normal dykes are numbered, there is a

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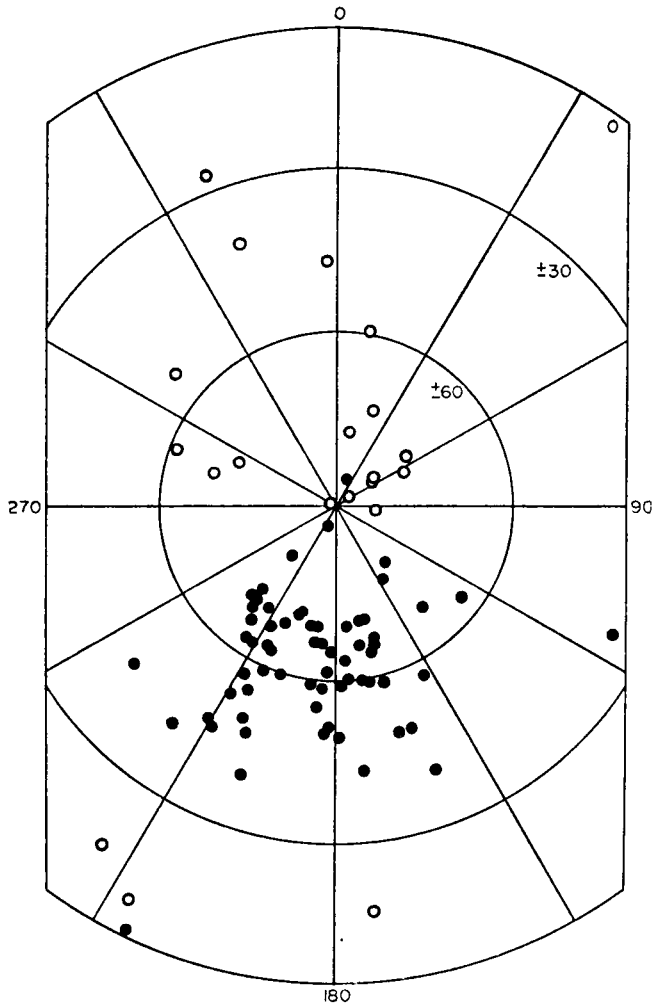


FIG. 2. Dyke-mean directions of magnetization for the 85 accepted dykes (out of 87). Dyke mean data are in the table in the Appendix. Solid symbols have upwards (negative) magnetizations.

suggestion that the intrusion of dykes progressed roughly from one end of the swarm to the other (the sense would not be known), during which time the normal field direction swung through a large loop beginning and ending at near-vertical inclination. However, one cannot put the suggestion to further proof, because of the lack of intersecting dykes to determine the sequence of intrusion of the individual dykes comprising the swarm.

In Fig. 1, the normally magnetized dykes are numbered on the landward side (to the east) of the coast, while the reversed dykes are numbered on the seaward side. There is no evidence that the normal dykes occur preferentially at one part of the swarm, so that we cannot say positively that the swarm did progress from one end to the other during (for example) an R-N-R history of the geomagnetic field. The greatest density of normal dykes occurs where the reversed dyke density is also greatest. It seems therefore likely that the reversed dykes were emplaced during one or several episodes, separately from the normal dykes, and that the density of emplacement along the swarm was controlled by the same local conditions during each episode.

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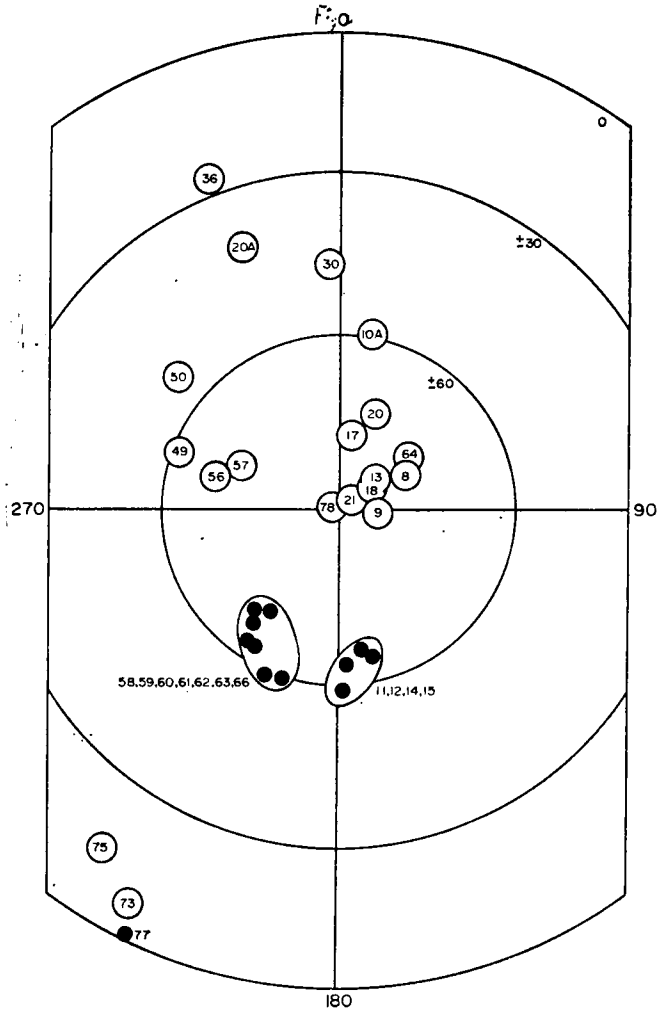


FIG. 3. All of the normal dykes are numbered, and show some tendency to following a large loop path as one progresses along the swarm. Three other ' dyke groups ' are displayed, which may represent near-simultaneous injection of nearly adjacent dykes. Solid symbols indicate upwards (negative) inclinations.

4.2. *Reversely magnetized dykes*

As for the Tertiary lavas of Antrim (Wilson 1970), the Skye lavas (Wilson *et al.* 1972), and the Mull dykes (Ade-Hall *et al.* 1972), we again find ' shallow reversed ' dykes, with low southerly inclinations of natural magnetization (Fig. 2). Although the meaning of this phenomenon is not clear, it has the following positive characteristics:

(a) It has so far been general to the British Tertiary igneous province.

(b) Many lines of argument (Ade-Hall *et al.* 1972) suggest that shallow reversed directions are not due to remagnetizations, but reflect the original field direction during dyke intrusion. The principal arguments rely on the coexistence of N and R dykes near to each other, and the agreement of baked rocks with their associated dykes' magnetizations. The opposite point of view is taken by Lovlie, Gidskehaug & Storetvedt (1972).

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4.3. *Dyke groups*

It has been found earlier (Doell 1969; Wensink 1970; Wilson 1970; Wilson *et al.* 1972) that a sequence of lavas in known stratigraphic order often contains a group (or groups) of adjacent lavas with very similar directions of magnetization. This stratigraphic order is not present in a dyke swarm. Nevertheless, it is perhaps worth looking for palaeomagnetic 'dyke groups' of closely adjacent dykes, because several fairly close dykes could have been injected simultaneously, or nearly so.

The normal dykes in the Vaternish suite (Section 4.1 and Fig. 3) may constitute a closely contemporaneous group. Apart from those, there seem to be three other dyke groups, shown in Fig. 3. They are:

group 1: dykes 11, 12, 14, 15

group 2: dykes 58, 59, 60, 61, 62, 63, 66

group 3: dykes 73, 75, 77 (all shallow reversed).

Their internal agreements suggest that any one of the three groups contains dykes intruded nearly contemporaneously, perhaps within one hundred years of each other, although the possibility of accidental agreement cannot be entirely excluded, nor can unusually long periods of low secular variation.

4.4. *Statistics*

The overall results (Fig. 2) show such a non-standard distribution of directions, that we do not feel that a mean pole position should yet be considered (as with the Skye lavas, Wilson 1972). We prefer to await more results from the British Tertiary Province, particularly those from work in progress on about 800 dykes from Skye and Arran, before overall interpretations are attempted.

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Appendix

Dyke	$N(N)$	Decl	Incl	α_{95}
A03I000	4(4)	221·5	−69·7	4·65
A03I001	4(4)	181·4	−64·9	4·16
A03I002	3(4)	125·8	−63·4	3·09
A03I003	4(4)	174·3	15·9	4·62
A03I004	3(4)	184·4	−58·5	3·30
A03I005	3(4)	208·0	−57·4	4·72
A03I006	3(4)	203·6	−63·1	4·01
A03I007	4(4)	175·4	−60·2	3·86
A03I008	4(4)	63·1	77·0	3·79
A03I009	3(4)	95·4	83·3	4·88
A03I010	4(4)	201·8	−48·2	6·99
A03I010A	4(4)	10·8	59·6	4·13
A03I010B	4(4)	220·6	−78·9	2·73
A03I011	4(4)	163·8	−65·0	4·95
A03I011A	4(4)	160·9	−49·3	2·56
A03I012	4(4)	176·0	−63·7	2·71
A03I013	4(4)	52·9	82·0	2·66
A03I014	4(4)	169·8	−65·9	3·68
A03I015	4(4)	177·7	−58·8	1·62
A03I016	4(4)	197·2	−86·5	3·13
A03I016A	4(4)	114·9	−36·7	3·30
A03I017	5(5)	10·6	77·7	2·81
A03I018	4(4)	52·9	87·3	5·76
A03I019	4(4)	21·9	−85·1	4·31
A03I019B	4(4)	205·7	−66·7	1·22
A03I020	4(4)	21·2	72·6	3·92
A03I020A	4(4)	339·9	41·7	3·90
A03I021	4(4)	53·9	82·2	7·25
A03I024	3(4)	164·5	−65·9	2·86
A03I025	3(4)	153·0	−57·4	5·50
A03I026	4(4)	139·3	−67·3	2·95
A03I027	4(4)	182·5	−50·5	4·04
A03I027A	4(4)	205·9	−64·8	2·34

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A03I028	4(4)	159·0	−40·6	2·74
A03I029	4(4)	197·8	−70·7	1·81
A03I030	4(4)	357·9	47·7	4·88
A03I031	4(4)	220·7	−71·4	2·29
A03I032	4(4)	187·5	−59·2	3·06
A03I033	4(4)	186·0	−66·4	4·56
A03I034	4(4)	165·1	−70·0	2·43
A03I035	4(4)	137·9	−77·2	3·51
A03I036	4(4)	338·6	26·7	3·99
A03I037	4(4)	203·2	−50·3	3·54
A03I038	4(4)	185·0	−55·2	2·77
A03I039	7(7)	219·2	−69·3	2·20
A03I040	2(4)	164·3	−58·6	3·48
A03I041	3(4)	167·0	−70·1	2·91
A03I042	4(4)	170·4	−59·7	4·17
A03I043	4(4)	182·4	−61·5	3·35
A03I044	4(4)	205·1	−55·0	1·80
A03I045	3(4)	163·9	−49·3	2·54
A03I046	4(4)	173·4	−43·4	2·45
A03I047	3(4)	181·4	−52·0	2·48
A03I048	4(4)	194·5	21·5	19·06*
A03I049	4(4)	289·7	61·4	3·49
A03I050	4(4)	309·6	54·6	3·78
A03I051	4(4)	291·6	37·9	31·80*
A03I052	4(4)	216·2	−42·9	4·03
A03I053	4(4)	231·3	−46·0	3·44
A03I054	4(4)	165·7	−64·3	2·05
A03I055	4(4)	169·4	−59·3	4·03
A03I056	4(4)	285·2	68·9	2·59
A03I057	4(4)	294·7	72·2	3·03
A03I058	4(4)	212·4	−69·3	2·25
A03I059	3(4)	215·5	−66·1	2·43
A03I060	4(4)	202·9	−59·3	1·82
A03I061	4(4)	218·8	−68·0	4·14
A03I062	3(4)	213·7	−63·0	2·37
A03I063	4(4)	210·7	−63·0	2·36
A03I064	4(4)	54·3	75·7	1·81
A03I065	4(4)	189·8	−69·1	4·29
A03I066	4(4)	197·3	−59·9	3·27
A03I067	4(4)	208·5	−53·1	2·13
A03I068	4(4)	186·2	−66·2	4·70
A03I069	4(4)	178·8	−49·9	5·63
A03I070	4(4)	189·6	−69·2	2·68
A03I071	4(4)	203·1	−68·2	1·74
A03I072	4(4)	174·3	−69·5	2·41
A03I073	4(4)	207·3	8·0	6·96
A03I074	4(4)	147·0	−75·2	2·64
A03I075	4(4)	214·0	15·4	3·30
A03I076	3(4)	197·7	−70·9	2·70
A03I077	4(4)	206·0	−1·7	3·72
A03I078	4(4)	291·4	89·1	2·15
A03I079	3(4)	199·0	−40·2	3·83
A03I080	4(4)	210·2	−47·1	1·84
A03I081	4(4)	208·9	−46·5	1·83

The mean direction of magnetization and α_{95} for each dyke in swarm I from Vaternish Peninsula, Skye.

Decl = declination of natural magnetization.

Incl = inclination of natural magnetization.

α_{95} = the half angle of the cone of confidence for the natural magnetization.

$N(N)$ = number of separately oriented samples used in anlysis (number collected).

Rejected dykes have an asterisk beside their α_{95} value.

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